

MEDIMATCH · NATIONAL REDISTRIBUTION COMMAND

See surplus. Detect need. Coordinate impact.

Leveraging geospatial intelligence
for equitable medical-supply redistribution

Lesnar Gitonga · USIU-Africa · 23 June 2026

Conference demo · synthetic inventory · no patient records

The problem is not only scarcity. It is disconnected visibility.

Facilities can hold usable surplus while others face urgent shortfalls. MediMatch makes both sides visible in one operating picture.

56.3%

face weekly or monthly stockouts

82.8%

rate manual methods ineffective or slow

57.8%

report wastage of valid supplies

89.1%

willing to pilot the system

A supply gap becomes a coordination question

Which available source is close enough, sufficiently stocked, verified, product-compatible—and practical to move now?

57%

KEMSA national order-fill rate,
as of mid-2025 (per the abstract)

The baseline is historical; the deck does not present it as the current 2026 rate.

We asked the people who run the supply rooms

64 healthcare professionals across Nairobi County · 80% response rate · 2025 field study

56.3%

experience stockouts of essential items on a weekly or monthly basis

82.8%

rate current manual, ad-hoc redistribution as “ineffective” or “slow”

57.8%

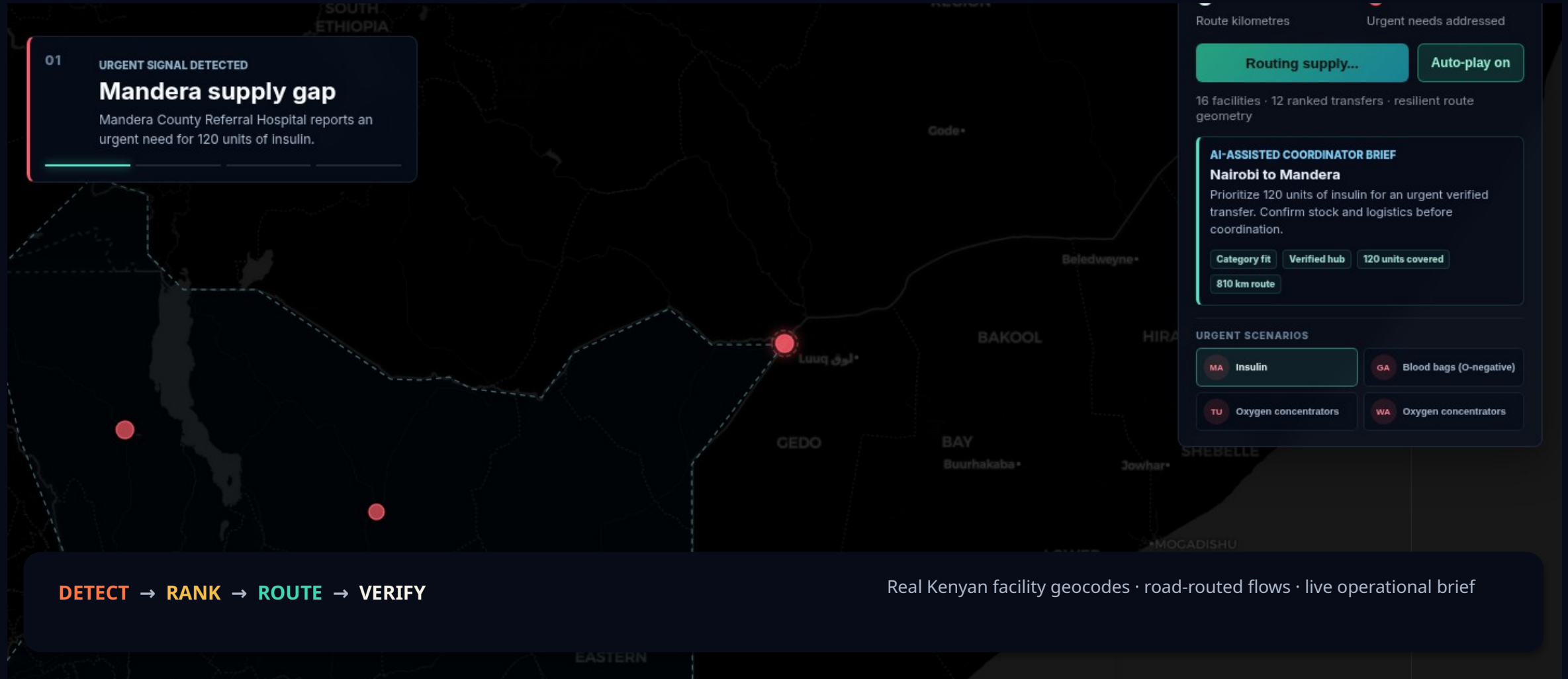
identify the wastage of valid medical supplies as a primary operational challenge

89.1%

are willing to pilot the MediMatch platform

Respondents span clinical staff (nurses & pharmacists), facility administrators, and pharmacy/inventory & technical personnel — a predominantly young, digitally-literate workforce.

From isolated reports to one national operating picture

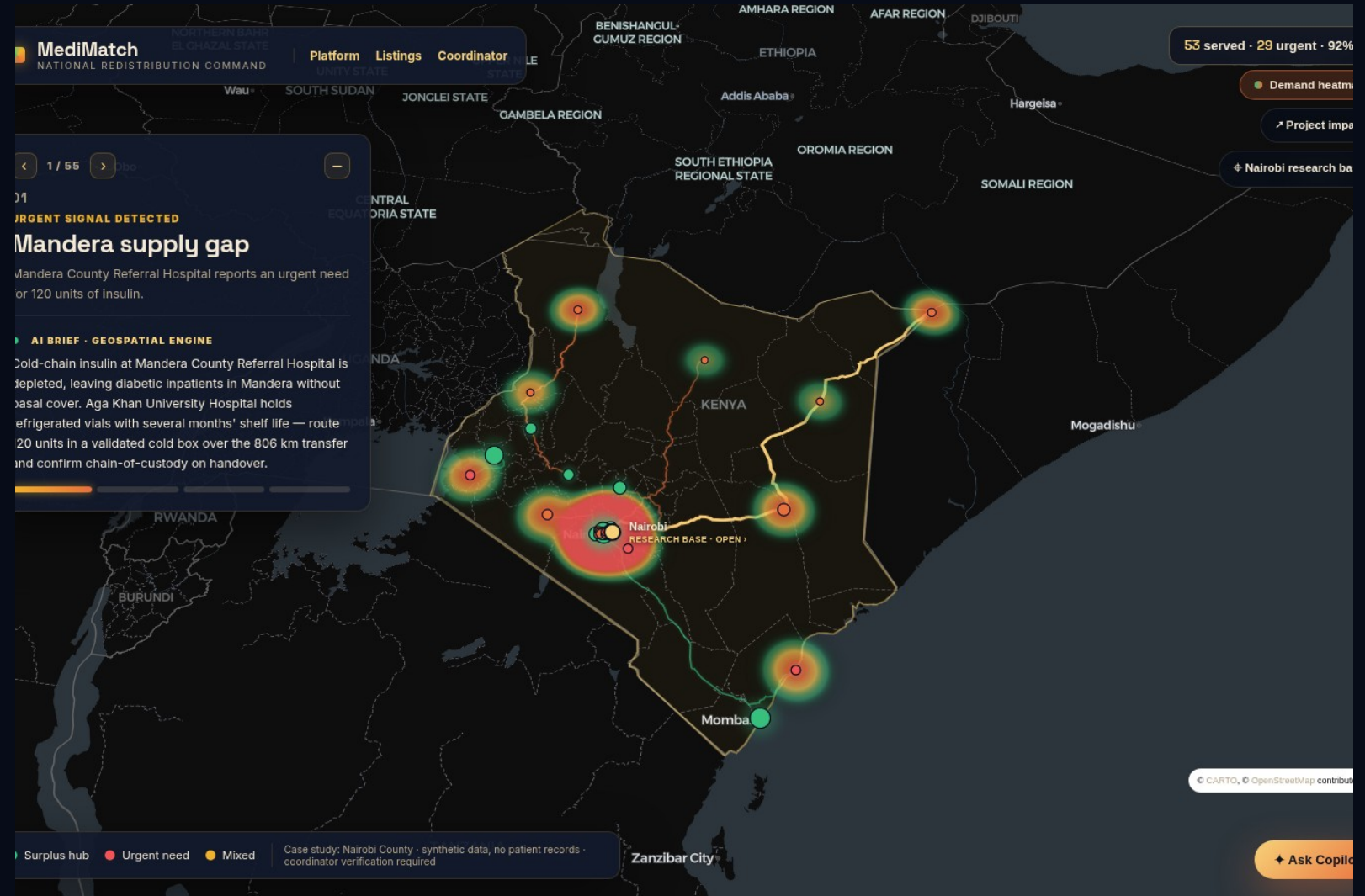


See demand concentration before routing

What geography adds

- Reveal where shortfall clusters—not just where the loudest request originated.
- Compare demand against available hubs and connected road corridors.
- Use distance as one signal alongside urgency, product fit and verification.
- Move from 'nearest' to 'best feasible match'.

Geography supports triage; it does not authorize a transfer.



Nairobi turns the national concept into a street-scale

workflow inside Nairobi County

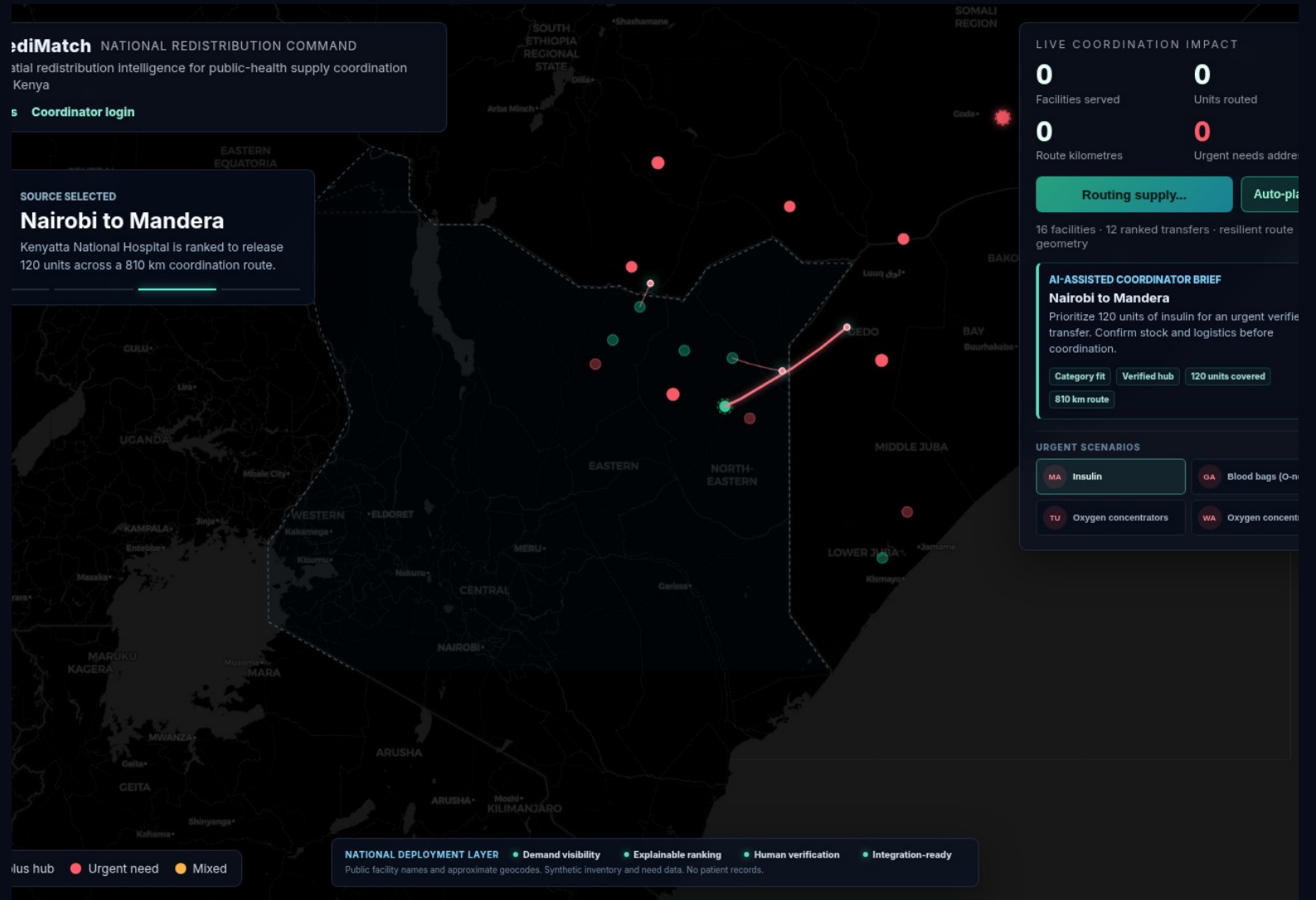


The system shows why a source was selected

Decision signals

- Proximity** direct and routed distance
- Urgency** clinical supply priority
- Product fit** category and cold-chain needs
- Quantity** full or partial coverage
- Verification** trusted source and ownership

Curated situation briefs explain the route in plain operational language.



What the current throughput could unlock over one year



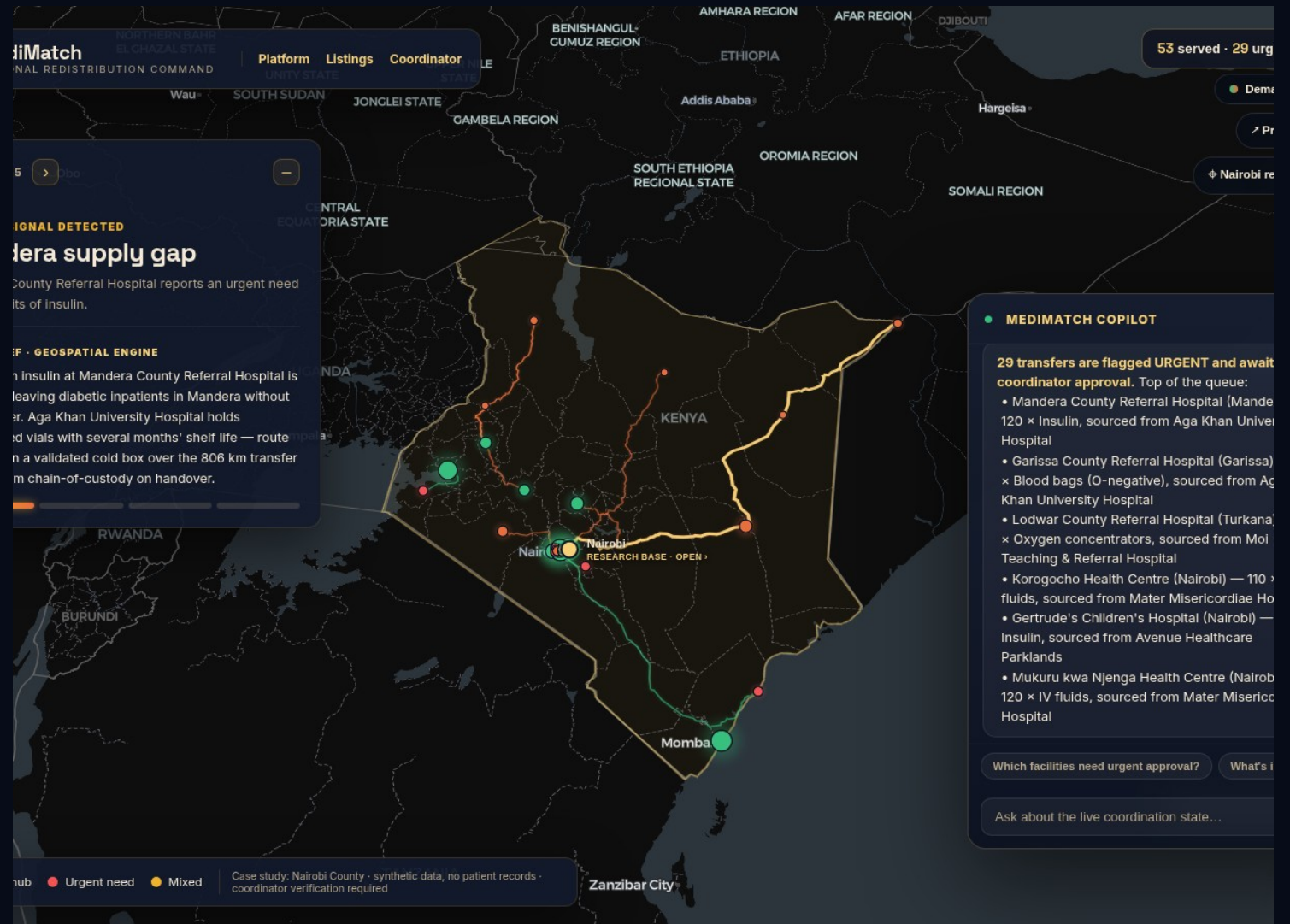
MODEL — NOT A GUARANTEE. Weekly cycles, 42% near-expiry recovery, coverage saturates as facilities onboard.

Copilot answers from the current operational state

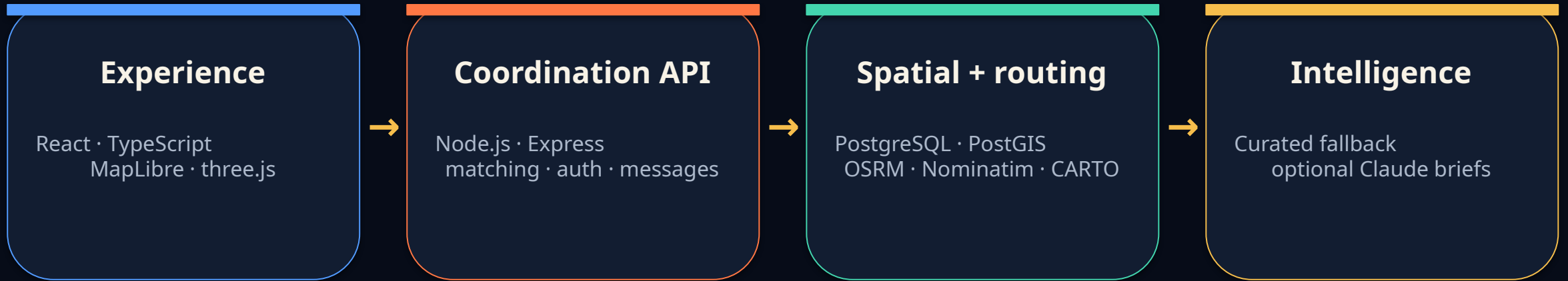
Ask operational questions

- Which facilities need urgent approval?
- What is in transit right now?
- Where are the shortfalls in Mandera?
- Who has surplus oxygen?
- Summarise the national situation.

Presentation-safe mode is grounded in live plan data and works without an API key. An Anthropic key can enable Claude-written briefs.



A production-minded spatial stack with a human decision boundary



Human verification is the final control

No diagnosis · no autonomous transfers · synthetic demo inventory · explicit assumptions · coordinator sign-off

Location is not just a field. It is a coordination advantage.

The live demo

- 01 Globe intro → Kenya command floor
- 02 Toggle the demand heatmap
- 03 Open the Nairobi research base
- 04 Project impact over one year
- 05 Ask Copilot for urgent approvals

MEDIMATCH

**See surplus. Detect need.
Coordinate impact.**

Questions?